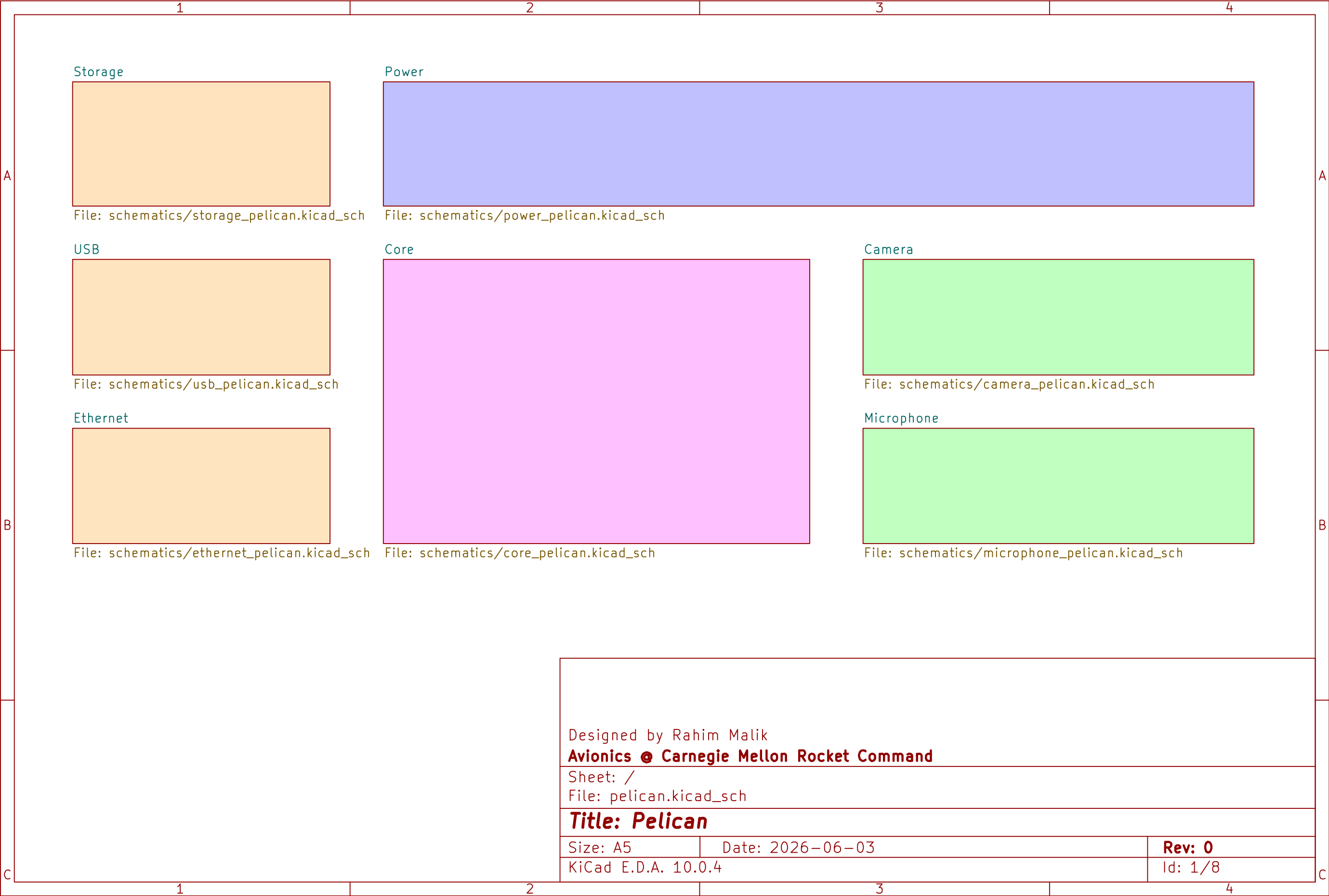




[illegible]

Battery Charging

R9 = KISET / ICHG KISET = 890, ICHG = 1A
R12 = KILIM / IIN_LIMIT KISET = 1610, ICHG = 1.5A

The diagram illustrates a battery charging circuit using the BQ24074RGTT IC. The IC is configured with the following components and connections:

- Power Supply:** VCC_3V3 and VCC_5V0.
- Resistors:**
 - R9 (1K10) connected to VCC_5V0 and the IN pin.
 - R12 (887R) connected to VCC_5V0 and the ITERM pin.
 - R11 (100K) connected to VCC_3V3 and the VSS pin.
 - R10 (10K) connected to the BAT pin and GND.
 - R35 (10K) connected to VCC_3V3 and the BAT_CHG_STAT pin.
- Capacitors:**
 - C4 (1uF) connected to VCC_5V0 and GND.
 - C5 (4.7uF) connected to the OUT pin and GND.
 - C6 (4.7uF) connected to the VBAT pin and GND.
- IC Pins and Connections:**
 - IN (13) connected to R9.
 - IILIM (12) connected to R9.
 - TMR (14) connected to R9.
 - ITERM (15) connected to R12.
 - ISET (16) connected to R12.
 - IC (4) connected to GND.
 - IE (5) connected to GND.
 - EN2 (6) connected to GND.
 - EN1 (7) connected to GND.
 - VSS (8) connected to R11.
 - PAD (9) connected to GND.
 - BAT (2) connected to R10.
 - BAT#2 (3) connected to GND.
 - TS (11) connected to GND.
- Status Outputs:**
 - BAT_CHG_SRC_STAT (7) connected to VCC_3V3.
 - BAT_CHG_STAT (10) connected to VCC_3V3.
- Other Components:**
 - U3 BQ24074RGTT (IC).
 - VBAT (VBAT pin) connected to C6.

Power Mux

The diagram illustrates a Power Mux circuit using a TPS2116DRLR (U2) as the main component. The circuit is powered by VCC_3V3 and PWR_SRC. The input side features a USB connection (VUSB) and a VBAT_GATE switch. The output side includes a boost converter (C11, C12) and a boost input (BOOST_IN). The circuit is designed to convert USB 4.3V to a required 1V threshold using a divider (R4, R5). The output is connected to the boost converter, which is powered by PWR_SRC. The circuit is labeled with component values: C7 (10uF), C9 (100nF), C8 (10uF), C10 (100nF), R4 (330K), R5 (100K), C11 (10uF), and C12 (100nF). The output is connected to the boost input (BOOST_IN) and the boost converter (C11, C12). The circuit is labeled with component values: C7 (10uF), C9 (100nF), C8 (10uF), C10 (100nF), R4 (330K), R5 (100K), C11 (10uF), and C12 (100nF). The output is connected to the boost input (BOOST_IN) and the boost converter (C11, C12).

Divder sized to convert USB 4.3V at switch to required 1V threshold

Power Regulation

The diagram illustrates a boost converter circuit for power regulation. The input is labeled BOOST_IN, which is connected to a capacitor C1 (4.7uF) to ground. The input voltage is then connected to the VIN pin (pin 3) of the TPS61023DRLR (U1). The EN pin (pin 2) is connected to the input through a resistor R1 (100K) and to ground (pin 4). The SW pin (pin 5) is connected to an inductor L1 (FHD4020S-1R0MT) and to the VOUT pin (pin 6). The VOUT pin is connected to the output voltage VCC_5V0 through a resistor R2 (768K) and to the FB pin (pin 1) through a resistor R3 (100K). The FB pin is also connected to ground. The output voltage VCC_5V0 is filtered by a capacitor C2 (22uF) to ground and a capacitor C3 (22uF) in parallel with the load.

Divder sized according to $R2 = (V_{out}/V_{ref} - 1) * R3$
 $V_{ref} = 580-680mV$, $V_{out} = 5V$, $R3 = 100K$

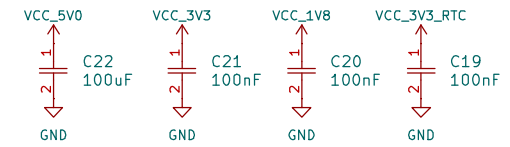
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Rev: 0
Id: 2/8

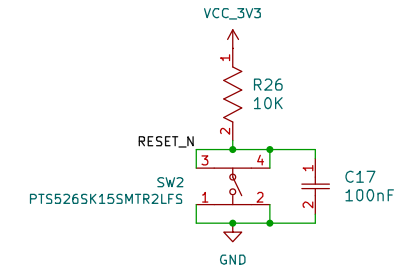
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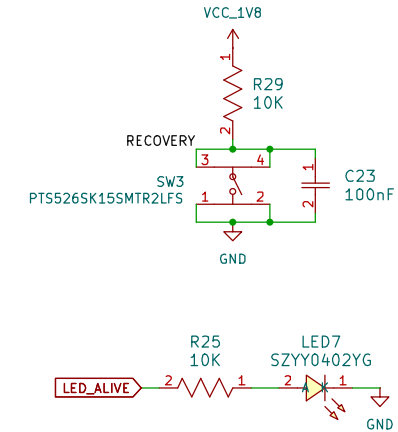
Decoupling and Brownout



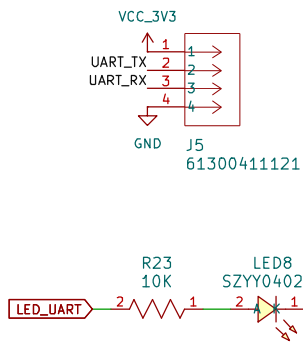
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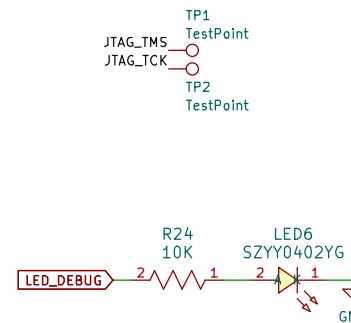
Program



UART



Debug

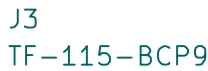


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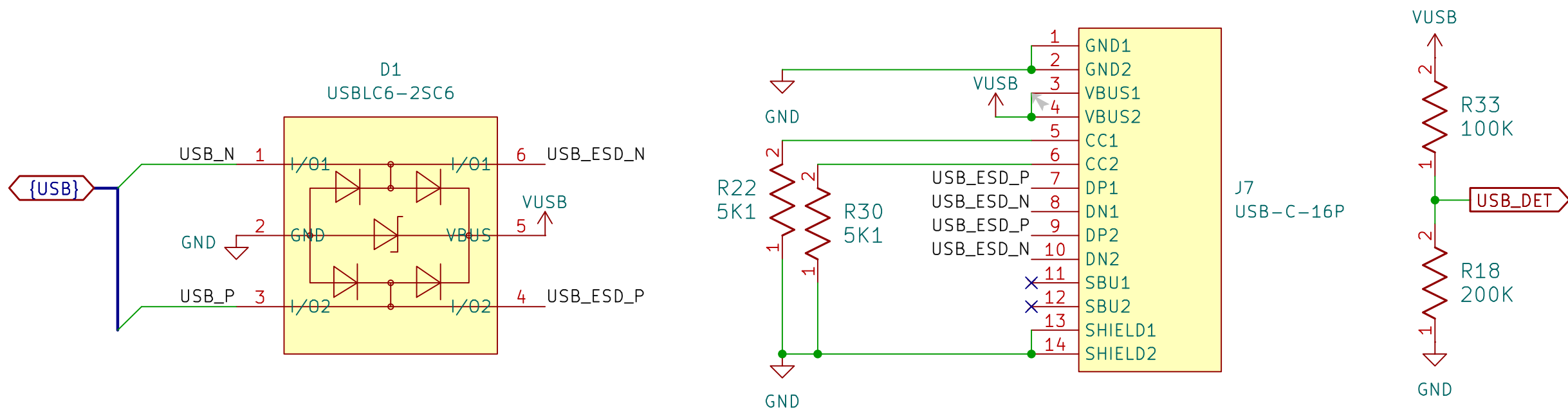
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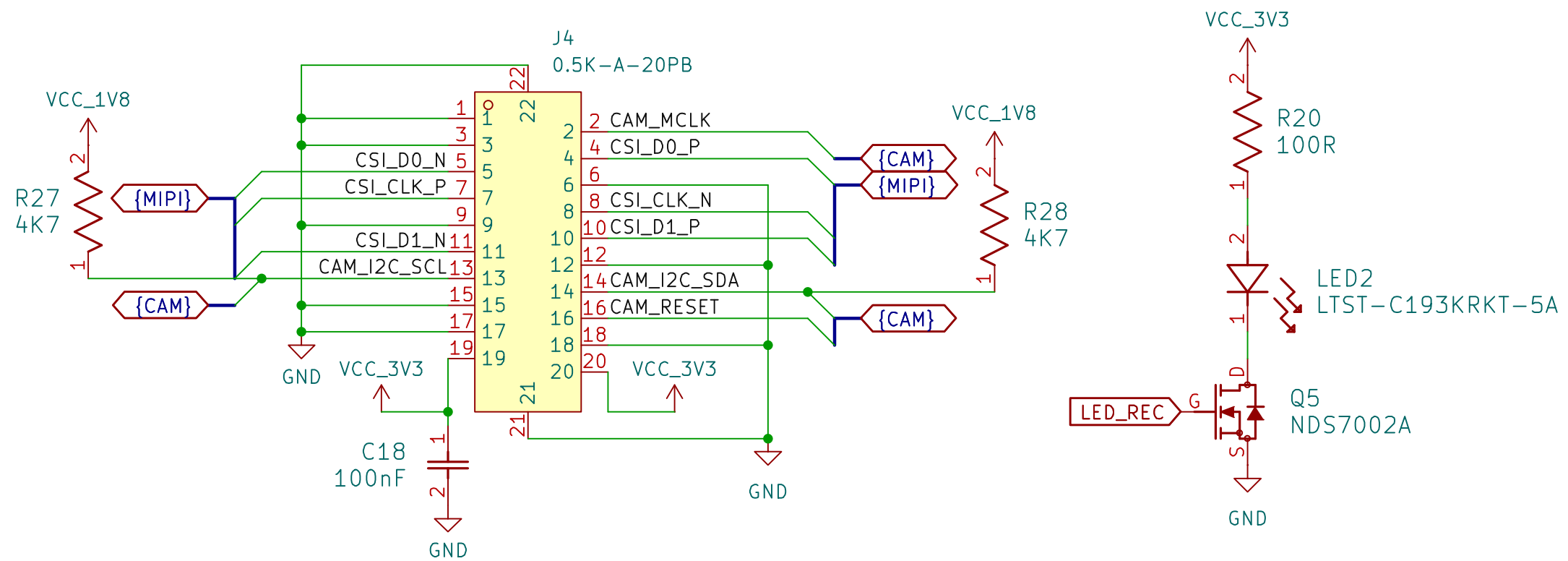
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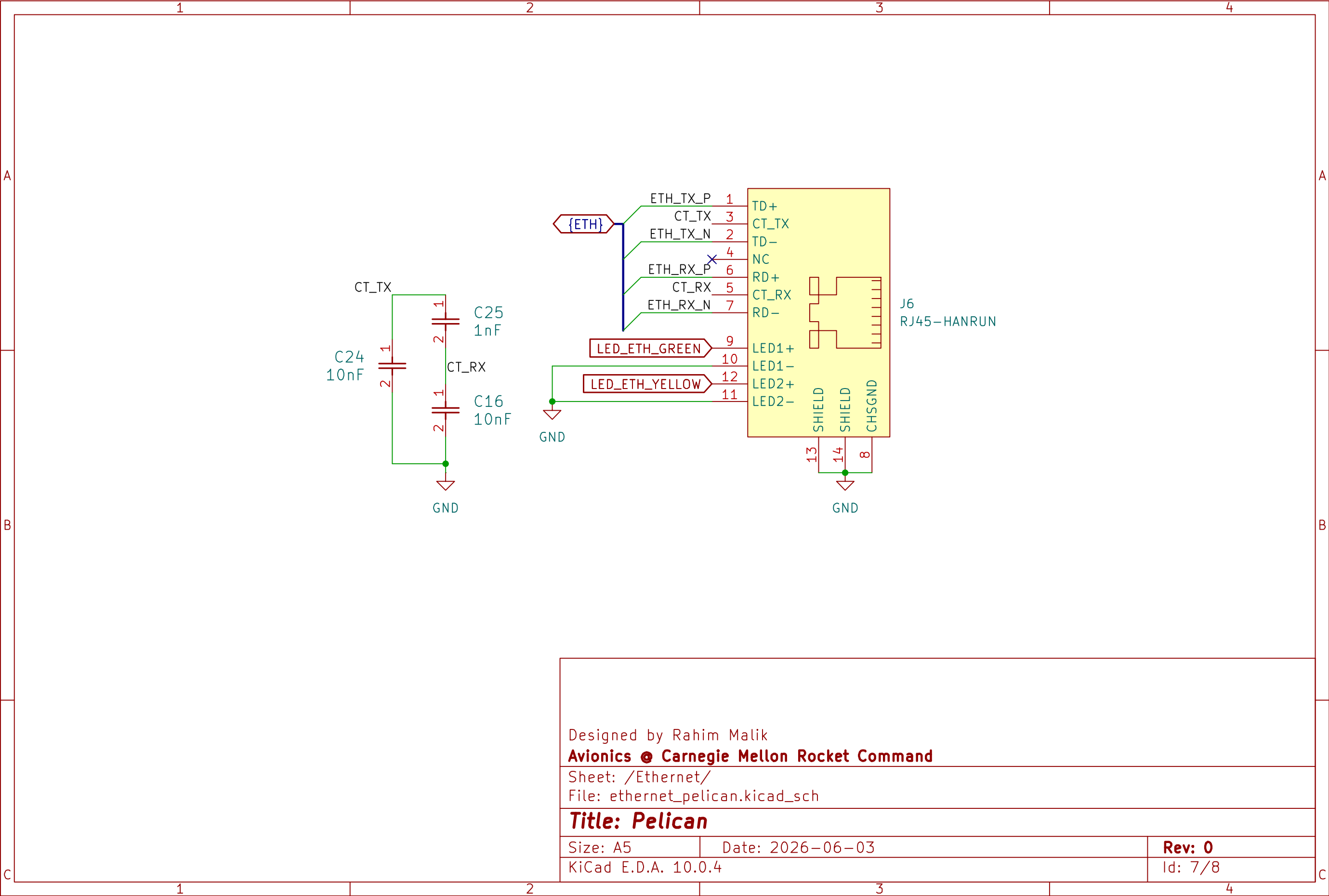
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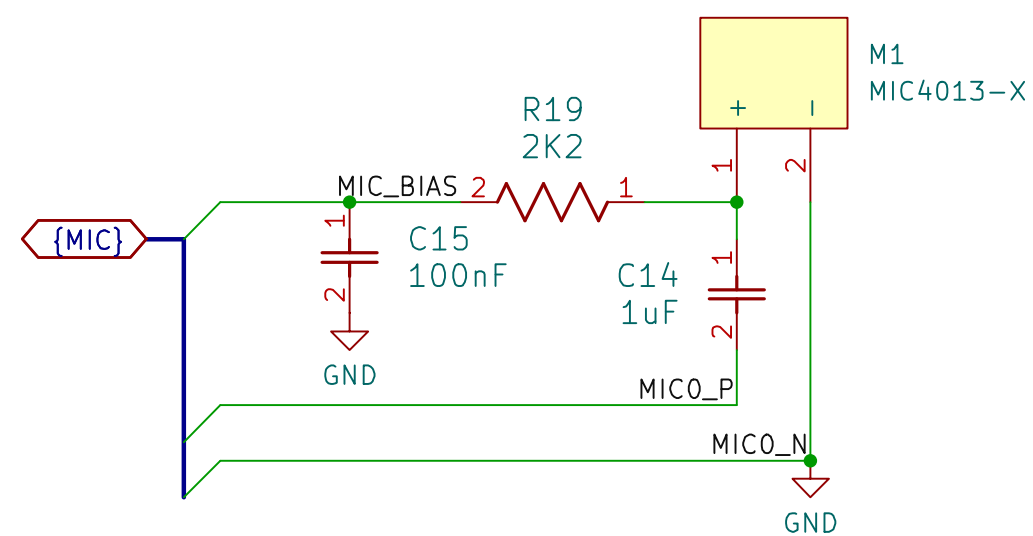
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Rev: 0

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